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G^{2}Net: A Geometry-Guided Network for Accurate Spine Morphology Estimation from a Single RGB Image of Human Back

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***Abstract*— Adolescent idiopathic scoliosis (AIS) assessment relies on quantitative characterization of spine morphology, however routine radiography exposes adolescents to ionizing radiation and is impractical for frequent screening and longitudinal monitoring. To address these limitations, this paper presents G^{2}Net, a geometry-guided framework for radiation-free and accurate spine morphology estimation from a single back-surface RGB image. G^{2}Net decomposes spinal morphology estimation into two stages. In the first stage, it performs high-fidelity back-surface depth estimation by jointly modeling global continuity and patch-wise local cues. In the second stage, it leverages the estimated depth map together with the RGB image as geometric and appearance conditions to generate a compact spine-curve map that represents spinal morphology. Experimental results demonstrate that G^{2}Net outperforms advanced monocular depth estimation baselines on widely used threshold-accuracy and error metrics. For spinal morphology estimation, the generated spine-curve maps show high agreement with X-ray–derived references, achieving an SSIM of 0.950 with low mean length and curvature differences of 0.079 and 0.028, respectively. These findings indicate that explicitly incorporating geometric priors improves robustness and anatomical consistency in spine morphology estimation, enabling scalable, radiation-free AIS assessment from a single RGB image at inference.**

***Index Terms*— Adolescent idiopathic scoliosis, radiation-free, geometry guidance, depth estimation, low-rank adaptation, spine morphology estimation.**

I. Introduction

ADOLESCENT idiopathic scoliosis (AIS) is a prevalent three-dimensional (3D) spinal deformity that primarily manifests during the adolescent growth spurt [1]. In addition to postural and aesthetic concerns, AIS may lead to persistent pain [2], cardiopulmonary compromise [3], and motor dysfunction [4], which can result in long-term adverse effects on physical and mental health and overall quality of life. Clinical screening and follow-up therefore rely heavily on quantitative assessment of spinal morphology. Radiographic imaging remains the diagnostic gold standard but repeated exposure to ionizing radiation together with high cost and limited portability restricts its suitability for large-scale screening and long-term monitoring [5], [6]. Radiation-free physical examinations such as the Adams forward bending test are accessible but often subjective and insufficiently sensitive, particularly for mild deformities, and may deviate from radiographic measurements [7]. These limitations highlight the need for accurate, scalable, and radiation-free solutions to support early detection and management of AIS [8].

With advances in deep learning and computer vision, optical approaches based on RGB imaging have gained increasing attention for radiation-free scoliosis assessment [9]. Representative studies include synthesizing radiographs from single RGB images using conditional generative adversarial networks(GANs) [10] and combining object detection with curve fitting to estimate curvature types and angles from smartphone photographs [11]. While these methods enable contactless and potentially real-time assessment, RGB images mainly encode two-dimensional appearance information and lack explicit geometric cues required to characterize underlying three-dimensional spinal anatomy. Consequently, RGB-only approaches are sensitive to illumination and shadow variations and often struggle to capture subtle morphological structures [12].

To compensate for missing geometric information, recent work has incorporated depth cues to enhance spinal structure

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analysis. Examples include depth-assisted learning for Cobb angle prediction [13], RGB and depth-based reconstruction of 3D spinal geometry from the back surface [14], and ultrasound and RGBD fusion for Cobb measurement and probe localization [15]. Although depth provides valuable geometric constraints, RGBD systems typically require additional sensors, calibration, and controlled acquisition conditions, which limits their scalability in screening scenarios. As a result, monocular depth estimation from a single RGB image has emerged as a promising alternative for practical and radiation-free 3D spinal assessment.

However, most existing monocular depth estimation(MDE) methods are designed for natural scenes with large depth ranges [16], [17]. In contrast, geometric variations of the human back surface usually fall within a centimeter-scale range, which demands substantially higher precision and sensitivity to subtle local undulations. Existing methods are not optimized for this small-range, high-accuracy setting, leading to insufficient modeling of fine-grained back-surface geometry. Our experiments further reveal a trade-off between local and global depth modeling. Models trained on local patches achieve higher regional accuracy but often introduce discontinuities near patches boundaries, whereas global depth estimation produces smoother and more consistent maps at the cost of oversmoothing local undulations, as illustrated in Fig. 1.

Motivated by these observations, we propose G^{2}Net, a Geometry-Guided Network that not only overcomes the limitations of conventional depth estimators in capturing subtle back-surface undulations, but also alleviates the lack of explicit 3D cues in existing RGB-based scoliosis assessment. Our framework infers spinal morphology in two stages: (i) back-surface depth estimation(BSDE) and (ii) spinal morphology estimation(SME). In the first stage, a shared image encoder jointly processes the full image and its patchified local regions to extract global and local multi-scale features. These features are progressively aggregated via a Mamba-based feature integrator, a multi-scale feature fusion module, and a cascade upsampling decoder to produce a high-fidelity back-surface depth map. In the second stage, we design a lightweight diffusion model that conditions on the surface image and the estimated depth to generate an accurate spinal curve map, while low-rank adaptation (LoRA) is employed to improve parameter efficiency and accelerate training.

Our contributions are summarized as follows :

- We propose G^{2}Net, a geometry-guided framework that integrates global and patch-wise multi-scale features, improving the accuracy and continuity of back-surface depth estimation while capturing subtle surface undulations.
- We design a LoRA-based multi-level geometric prior injection strategy that maintains anatomical fidelity of spinal curves, reduces model complexity, and accelerates training.
- We construct a large-scale, clinically curated multimodal dataset comprising paired RGB back images, depth maps, and full-spine radiographs from 2,213 adolescents, with expert-annotated ground truth spine curve maps to support reliable model training and evaluation.

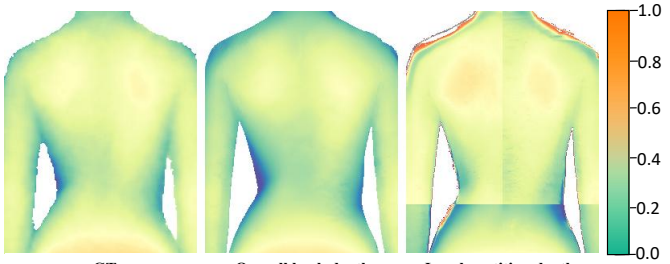


Fig. 1: Illustration of regional inconsistency in existing monocular depth estimation methods. From left to right, real depth, overall back depth estimation and depth estimation on each local partition. These results highlight the challenge of achieving both global continuity and local detail preservation when estimating centimeter-scale back-surface depth from RGB images.

- Experimental results show that G^{2}Net achieves competitive performance in back-surface depth estimation with $\varepsilon_{\mathrm{ARE}} = 0.053$ and $\varepsilon_{\mathrm{RMSE}} = 0.048$, outperforming many state-of-the-art (SOTA) methods. For spinal morphology estimation, our approach reaches $\mathrm{SSIM} = 0.950$ with low mean differences in length (0.079) and curvature (0.028), consistently surpassing advanced conditional generative models.

II. Related Works

A. Clinical assessment of scoliosis

In clinical practice, radiography remains the gold standard for diagnosing AIS due to its high accuracy in Cobb angle measurement [18]; However, concerns regarding ionizing radiation exposure, cost, and limited portability restrict its suitability for large-scale screening and frequent follow-up [19]. Radiation-free physical examinations, such as the Adams forward bending test [20], are therefore widely adopted in community and school settings. Despite their convenience, these assessments are subjective and exhibit limited sensitivity to mild deformities, often showing poor agreement with radiographic findings [21]. To bridge the gap between safety and objectivity, non-ionizing alternatives including surface topography and 2D photographic analysis have been explored [22]. Nevertheless, the lack of explicit 3D geometric modeling limits their ability to capture true spinal deformity, particularly in early-stage AIS [23]. These challenges motivate the development of learning-based automated methods for more objective and scalable AIS assessment.

B. Learning-based scoliosis assessment

Learning-based methods have enabled increasingly objective and automated scoliosis assessment. Early studies mainly relied on end-to-end regression, where CNNs with ResNet-style backbones directly predicted Cobb angles from radiographs [24]. Despite promising accuracy, such models are sensitive to noise, projection changes, and curve variability, which limits robustness in clinical practice [25].

To improve stability and interpretability, landmark-based pipelines were subsequently introduced. These approaches typically employ U-Net or HRNet architectures to localize vertebral landmarks, followed by geometric fitting for Cobb angle computation [26], [27]. By explicitly modeling anatomical structure, landmark-based methods generally produce more robust and structured outputs that support downstream clinical analysis [28].

Driven by the clinical demand for low- or zero-radiation assessment, learning-based methods have further expanded to optical modalities, particularly RGB back-surface images. Representative approaches combine CNN-based detection or attention mechanisms with curve fitting to estimate spinal deformity from monocular images [29]. However, RGB-only methods lack explicit three-dimensional geometric constraints, making them sensitive to viewpoint and illumination variations and less reliable for moderate-to-severe deformities [30].

Incorporating depth information provides direct geometric cues for three-dimensional modeling. Zhao *et al.* [31] proposed LatXGen, a dual-stage RGBD framework that synthesizes lateral spinal radiographs from back-surface data to enable radiation-free sagittal alignment analysis. Other RGBD-based approaches similarly leverage depth maps or point-cloud representations with CNNs to improve robustness over RGB-only methods [32]. Nevertheless, the reliance on dedicated RGBD hardware and calibration limits scalability in unconstrained screening scenarios, motivating the recovery of reliable depth information from monocular RGB images.

C. Monocular depth estimation

MDE aims to infer scene depth from a single image. Over the past decade, CNN-based architectures have dominated this task, commonly adopting encoder–decoder designs and multi-scale feature fusion to improve prediction accuracy [33]. However, due to their inherently local receptive fields, CNNs struggle to model long-range dependencies, which limits global contextual reasoning and constrains further performance gains [34].

To enhance global modeling, Transformer-based architectures incorporating self-attention have been introduced into MDE. Representative approaches combine Transformers with convolutional features to capture both long-range context and local geometry, achieving improved accuracy [29], [35]. Nevertheless, the quadratic computational and memory cost of self-attention makes such models inefficient for high-resolution depth estimation.

Recent efforts have explored hybrid convolutional designs inspired by Transformers, such as ConvNeXt, to balance efficiency and representation capacity [36]. However, prior studies indicate that convolution-based models remain limited in capturing global consistency and are often optimized for datasets with large depth ranges, making them less suitable for scenarios requiring fine-grained depth estimation [37], [38].

To address both efficiency and long-range modeling, Gu *et al.* [39] proposed Mamba, a state-space model–based architecture that captures long-range dependencies with linear complexity. By leveraging selective state-space representations, Mamba provides stronger global modeling than CNNs while being significantly more efficient than Transformer-based methods for high-resolution inputs.

D. Generative methods for image synthesis

Image synthesis learns data distributions to generate images consistent with real samples and remains a core topic in medical image analysis [40]. Early research was driven by GANs [32]. While variants like StyleGAN [41] improved training stability and realism in tasks such as cross-modality translation [42], GAN-based methods often suffer from mode collapse and struggle to preserve global anatomical coherence at high resolution.

Diffusion models have recently emerged as a powerful alternative with superior perceptual quality [43]. Beyond the foundational DDPM [44], various studies have significantly enhanced efficiency and controllability through accelerated sampling, latent space optimization, and large-scale pre-training [45]–[47]. To incorporate structural guidance, ControlNet [48] adds trainable branches to the diffusion backbone. However, these additional modules increase computational demands and require task-specific retraining, which reduces flexibility in clinical scenarios.

LoRA [49] offers a parameter-efficient alternative for adapting large generative models. Despite its efficiency, LoRA-based methods have limited capability to precisely control fine-grained geometric structures such as vertebral alignment and spinal curvature. This limitation motivates the development of geometry-aware generation frameworks tailored to clinical applications.

III. Method

An overview of the proposed G^{2}Net is shown in Fig. 2. The framework consists of two stages, namely 1) back-surface depth estimation (BSDE) stage that estimates the depth map of the unclothed back, and 2) spinal morphology estimation (SME) stage that infers the spine curve. Given an input color (RGB) image $I_c \in \mathbb{R}^{H\times W\times 3}$, the objective of the first BSDE stage is to estimate the depth map $\hat{D} \in \mathbb{R}^{H\times W}$, where H and W represent the image height and width, respectively. The subsequent SME stage takes both the estimated depth map $\hat{D}$ and the original RGB image I^c as inputs to generate a spine curve map $\hat{M}$ for spinal morphology estimation. In the following, we delve into the details of the major components of G^{2}Net. Specifically, Sections III-A and III-B describe the model design and implementation for the BSDE stage and SME stage, respectively.

A. Back-surface depth estimation

Compared with monocular depth estimation in natural scenes, where depth spans a wide range and strong parallax cues are often available, the human back exhibits a narrow depth range and subtle geometric undulations. In this setting, small depth errors may substantially distort downstream morphology inference. To better capture these fine-scale variations, we design a dual-branch architecture that extracts features from both the full image and its patchified local regions,

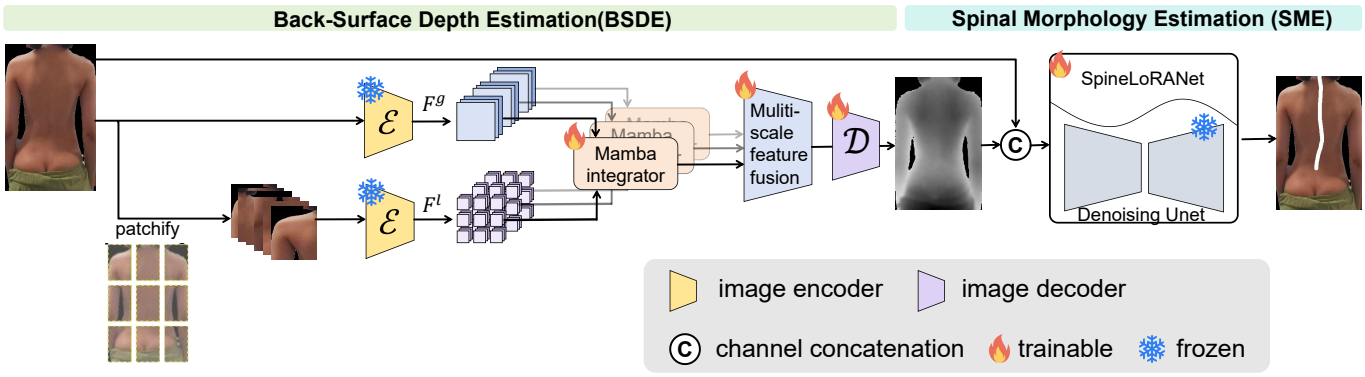


Fig. 2: Overview of the proposed G^{2}Net. The model comprises two stages: 1) the back-surface depth estimation (BSDE) stage, and 2) the spinal morphology estimation (SME) stage, along with a detailed integration of the entire model.

followed by collaborative modeling and cross-scale feature interaction. As shown in Fig. 2, the BSDE stage comprises four core components: 1) a ConvNeXt-based image encoder, 2) a feature integrator based on Mamba framework, 3) a cross-scale feature fusion module, and 4) a cascaded upsampling decoder.

1) ConvNeXt-based image encoder: We employ a pre-trained ConvNeXt-based image encoder for initial feature extraction, due to its computational efficiency and ability to capture both long- and short-range spatial dependencies. Specifically, we use the basic ConvNeXt backbone [32] as our image encoder $\mathcal{E}$ to extract a set of multi-scale global feature maps $\mathbf{F}^g$ from the input image I^c:

$$\mathbf{F}^g = \{F_i^g\}_{i=1}^n = \mathcal{E}(I^c), \tag{1}$$

where n denotes the number of output scales and in this work is set to 3. To enhance sensitivity to localized surface details, we further apply a patchify operator $\mathcal{P}(\cdot)$ (as popularized in ViT-style processing [50]). The image is partitioned into N non-overlapping patches $\{I_k^p\}_{k=1}^N$:

$$\{I_k^p\}_{k=1}^N = \mathcal{P}(I^c), \tag{2}$$

with patch size (h, w) and $N = \lceil H/h \rceil \cdot \lceil W/w \rceil$. H and W denote the height and width of the input image I^c and the operator $\lceil \cdot \rceil$ represents the ceiling function. Each patch is encoded using the same shared encoder $\mathcal{E}(\cdot)$ in parallel, and the resulting features are rearranged back to the original spatial layout to form multi-scale local feature maps:

$$\mathbf{F}^l = \{F_i^l\}_{i=1}^n = \mathcal{E}(\{I_k^p\}_{k=1}^N). \tag{3}$$

The global and local feature maps ($\mathbf{F}^g$ and $\mathbf{F}^l$) are subsequently fed into the subsequent feature integrator for further processing.

2) Mamba-based feature integrator: Addressing the challenge of inadequate cross-context interaction between macro-scale global structural cues and fine-scale local surface undulations, as well as inconsistent statistical properties that hinder stable fusion at each resolution level, we propose a targeted strategy. For the i-th scale, we first project the global feature map F_i^g and the local feature map F_i^l into a unified embedding space and concatenate them to form a joint sequence:

$$S_i = \left[\mathcal{F}\left(F_i^g\right),\ \mathcal{F}\left(F_i^l\right)\right], \tag{4}$$

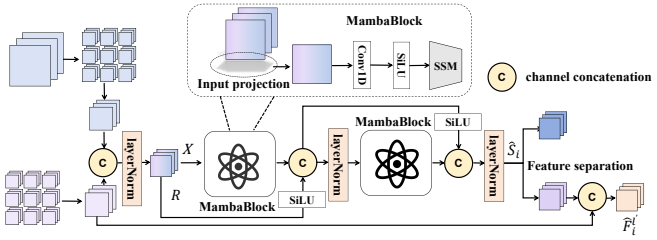


Fig. 3: Detailed framework of the Mamba-based feature integrator, where the SSM module adopts the Structured State Space Model formulation derived from classical State Space Model theory. [51]

where $\mathcal{F}(\cdot)$ denotes the linear projection operation.

To harmonize the statistical properties of global and local representations and ensure stable fusion, we normalize the joint sequence using Layer Normalization [52].

$$\tilde{S}_i = \mathrm{LN}(S_i) = \gamma \frac{S_i - \mu}{\sqrt{\sigma^2 + \epsilon}} + \beta. \tag{5}$$

The normalized sequence $\tilde{S}_i$ is then processed by a stack of dynamic fusion modules, i.e., MambaBlocks. Each block first applies an input projection to produce a main stream X and a residual stream R, followed by a depthwise 1D convolution to capture local dependencies. Long-range dependencies are modeled by a selective state space model (SSM) with the following recurrence:

$$h_t = e^{\Delta_t A} h_{t-1} + \Delta_t B x_t', \quad y_t = C h_t + D x_t', \tag{6}$$

where Δ_t denotes the learnable time step size, employed to regulate the dynamic propagation rate. A, B, C, and D are SSM parameters. The SSM output is gated by the residual branch and added back to the input through a residual connection, which stabilizes optimization and preserves information flow in deep stacks.

To balance local detail and global structure, we compute the final enhanced local feature $\hat{F}_i^l$ via an adaptive fusion:

$$\hat{F}_i^l = \sigma(a_i)\tilde{F}_i^l + (1 - \sigma(a_i))F_i^l, \tag{7}$$

where $\sigma(\cdot)$ denotes the sigmoid function that outputs a weight in the range $[0, 1]$, and a_i is a learnable scalar weight that

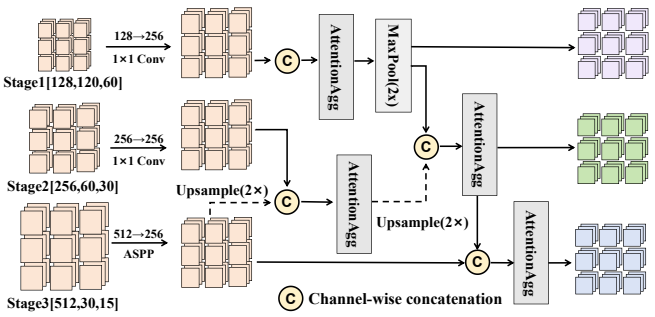


Fig. 4: Overview of the Cross-scale Feature Fusion module, where ASPP (Atrous Spatial Pyramid Pooling) is a spatial pyramid module composed of parallel atrous convolutions with different dilation rates for multi-scale representation. [53]

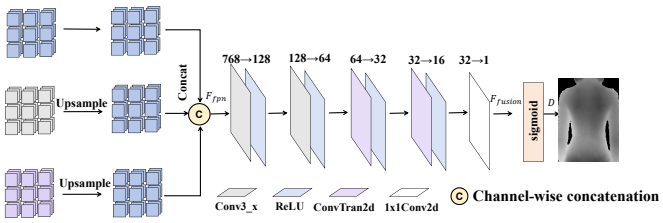


Fig. 5: Detailed framework of the cascade upsampling decoder module

dynamically adjusts the contribution of the Mamba-fused feature $\tilde{F}_i^l$ and the original local feature F_i^l.

3) Cross-scale feature fusion module: To further enhance feature representation and promote stable bidirectional interaction across scales, we design a cross-scale feature fusion module, as illustrated in Fig. 4. This module enhances the representational capacity of the obtained multi-scale features using the attention aggregation mechanism [54] and spatial pyramid pooling [55] to synergistically strengthen feature representation. Given the integrated multi-scale features $\{\hat{F}_i^l\}_{i=1}^3$, we first align channel dimensions using 1×1 convolutions:

$$L_i = \mathrm{Conv}_{1\times1}(\hat{F}_i^l), \quad (i = 1, 2, 3). \tag{8}$$

To enhance feature saliency during multi-scale fusion, we introduce a channel-spatial attention aggregation mechanism. Let $\mathcal{A}(\cdot)$ denote the attention aggregation operator, which integrates channel attention and spatial attention. Specifically, channel attention $CA(x)$ and spatial attention $SA(x)$ are formulated as:

$$CA(x) = \sigma\left(\mathbf{W}_2\delta\left(\mathbf{W}_1\mathrm{AvgPool}(x)\right)\right), \tag{9}$$

$$SA(x) = \sigma\left(\mathrm{Conv}_{7\times7}\left([\mathrm{avg}(x), \max(x)]\right)\right), \tag{10}$$

where $\delta(\cdot)$ denotes the ReLU activation function and $\sigma(\cdot)$ is the sigmoid function. The attention aggregation operation is defined as:

$$\mathcal{A}(x) = CA(x) \odot x + SA(x) \odot x, \tag{11}$$

with $\odot$ indicating element-wise multiplication.

To enable effective cross-scale feature interaction, a bidirectional spatial pyramid is constructed for multi-scale feature fusion. In the top-down pathway, high-level semantic features are progressively upsampled and injected into lower-level features, producing an intermediate top-down representation T:

$$T_{i-1} = \mathcal{A}\left(L_{i-1} + \mathrm{Up}(L_i)\right), \tag{12}$$

where $\mathrm{Up}(\cdot)$ denotes a $2\times$ upsampling operation. This process allows high-level contextual information to guide lower-level representations.

Subsequently, in the bottom-up pathway, structural and geometric information from lower scales is propagated back to higher scales by downsampling the intermediate features, compensating for missing local details in high-level semantics:

$$B_i = \mathcal{A}\left(T_i + \mathrm{Down}(T_{i-1})\right), \tag{13}$$

where $\mathrm{Down}(\cdot)$ represents a downsampling operation. Finally, the refined multi-scale feature pyramid $\{P_1, P_2, P_3\}$ is obtained from the bottom-up fused features at each corresponding scale, i.e., $P_i = B_i$ for $i \in \{1, 2, 3\}$. This bidirectional fusion strategy preserves both global contextual information and local geometric structures, yielding multi-scale representations for subsequent decoding.

4) Cascade upsampling decoder module: To reconstruct the final high-resolution depth map, we adopt a cascade upsampling decoder. We first upsample all pyramid features to the highest resolution and concatenate a fused feature map F_{fuse}, i.e.,

$$F_{fuse} = \mathrm{Concat}(Up(P_1), Up(P_2), P_3). \tag{14}$$

A 3×3 convolution with activation reduces channels and forms the initial decoded representation:

$$F_0 = \delta(\mathrm{Conv}_{3\times3}(F_{fuse})). \tag{15}$$

A sequence of strided transposed convolutions is then applied to progressively restore resolution:

$$F_k = \delta(\mathrm{ConvTran2d}_{s=2}(F_{k-1})), \quad k \in \{1, 2, 3\} \tag{16}$$

Finally, a 1×1 convolution with sigmoid normalization predicts the single-channel depth map $\hat{D}$, that can be formulated as,

$$\hat{D} = \sigma(\mathrm{Conv}_{1\times1}(F_k)). \tag{17}$$

B. Spinal morphology estimation

The spinal morphology estimation (SME) stage aims to generate a spine curve map $\hat{S}$ from the original RGB image I^c and the estimated back-surface depth map $\hat{D}$. G^{2}Net adopts a conditional diffusion formulation in which appearance and geometry are processed through decoupled yet tightly coupled signal flows. Specifically, the RGB image drives the generative diffusion process, while the depth map provides geometry-aware guidance that continuously modulates the diffusion trajectory.

The RGB image I^c is first fed into a frozen diffusion U-Net, which serves as the appearance-driven backbone. Following the standard reverse diffusion process, the backbone progressively transforms noisy latent variables into structured feature representations across multiple resolutions, forming a

hierarchical feature pyramid $\{h_i\}$. These features encode the visual context necessary for spine curve generation and remain unaltered by direct parameter updates during training.

In parallel, the estimated depth map $\hat{D}$ initiates a geometry-guided control signal flow. It is encoded by a lightweight prompt encoder $\mathcal{E}_\phi$ into multi-scale geometric features,

$$H_{\text{hint}} = \mathcal{E}_\phi(\hat{D}), \tag{18}$$

which capture global spinal alignment as well as local curvature cues inferred from the back-surface geometry. The encoded features are propagated through a ControlNet-style control branch that mirrors the hierarchical structure of the diffusion backbone.

To achieve parameter-efficient conditioning, each control block is implemented using LoRA-based residual layers, while the original backbone weights remain frozen. For a convolutional weight matrix W_0, the effective weight is expressed as

$$W = W_0 + \alpha BA, \tag{19}$$

where A and B are learnable low-rank matrices and α is a scaling factor. Timestep embeddings are injected into each block, ensuring that geometric guidance is synchronized with the diffusion stage.

As the geometry-guided features propagate through successive downsampling, bottleneck, and upsampling stages, the control branch produces a set of multi-resolution control features $\{c_i\}$. Each control feature is generated via a zero-initialized convolution, guaranteeing that the control pathway has no influence on the diffusion backbone at initialization and enabling stable training. During reverse diffusion, these control features are additively injected into the corresponding backbone features,

$$h_i' = h_i + c_i, \tag{20}$$

allowing geometric priors derived from $\hat{D}$ to modulate the diffusion process at every resolution.

Conditioned on the geometry-modulated feature hierarchy $\{h_i'\}$, the frozen diffusion decoder synthesizes the final spinal morphology map,

$$\hat{S} = \eta(\mathbf{h}'), \tag{21}$$

where $\eta(\cdot)$ denotes the reverse diffusion process. Through this continuous data flow—from appearance-driven diffusion to geometry-guided multi-level modulation—G^{2}Net effectively translates back-surface geometry and visual context into an anatomically consistent spine curve representation.

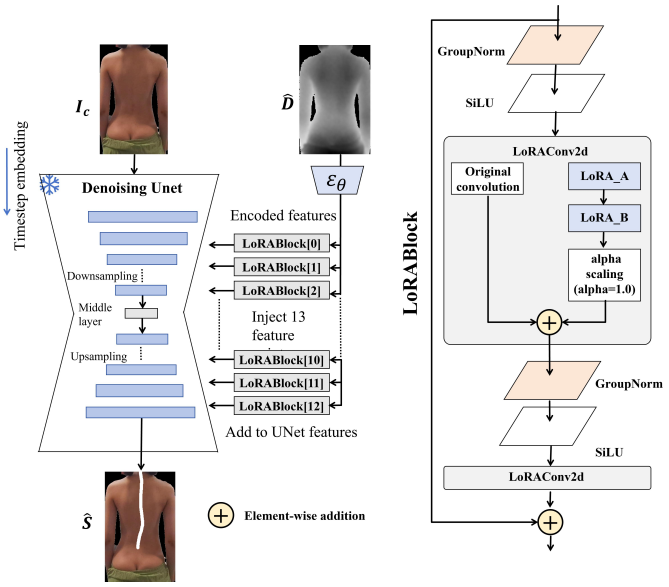


Fig. 6: Detailed framework of the G^{2}Net model. The LoRABlock approximates weight updates by introducing two low-rank matrices, LoRA-A and LoRA-B. LoRA-A projects the input into a low-dimensional space, whilst LoRA-B projects it back to the original dimension, thereby enabling parameter-efficient adaptation to downstream tasks [49].

IV. Experiments

A. Datasets

The dataset for this study was sourced from Queen Mary Hospital (QMH) and Duchess of Kent Children's Hospital (DKCH) in Hong Kong. All protocols were approved by the ethics review committees of both institutions (UW 15-596 and UW 25-300). The study included 2,213 adolescent participants aged 10–18 years, with a mean age of 16.02 years. Among them, 1,619 were female (73.16%) and 594 were male (26.84%). All participants (or their guardians) signed written informed consent forms prior to data collection. For each participant, data collection included one unclothed back RGB image, one depth image, and one standing full-spine posteroanterior radiograph covering the entire spine region. Depth images were captured using an Azure Kinect DK camera, while radiographs were acquired via standard hospital radiology equipment. Participants were excluded if they had psychological or neurological disorders, congenital physical deformities, prior spinal surgery, posture/motor dysfunction due to trauma, or concurrent tumor-related pathologies.

The dataset was stratified and randomly divided into training, validation, and test sets at an 8:1:1 ratio. For the BSDE task, depth images captured by the Azure Kinect DK served as GT. For the SME task, full-spine radiographs provided served as the clinical reference. To obtain pixel-level supervision for spinal morphology generation, we constructed a GT spine curve map from each radiograph as follows: (i) two senior spine surgeons jointly annotated four vertebral corner points per vertebra; (ii) vertebral centers were computed geometrically; (iii) the centerline was densified via cubic spline interpolation and smoothed with a Gaussian filter; and (iv) a continuous curve was rendered as a white line on a black background by drawing small disks (diameter 5 pixels) at sampled points and connecting adjacent segments.

During data preprocessing, pixel-level registration of RGB and depth images was performed using Azure Kinect SDK 1.4.1. Radiographs were aligned to the RGB images based on expert-annotated anatomical landmarks.

B. Evaluation Metrics

1) Back-surface depth estimation: We report standard depth estimation metrics, including *threshold accuracies* and *error*

measures.

a. Threshold accuracy. Following standard practice in monocular depth estimation, we report threshold-based accuracies δ_1, δ_2, and δ_3, which measure the fraction of pixels whose predicted depth is within a multiplicative error bound of the GT. For each valid pixel i, let $\widehat{D}_i$ and D_i denote the predicted and GT depth values. We define the per-pixel ratio:

$$\delta_i = \max\left(\frac{\widehat{D}_i}{D_i}, \frac{D_i}{\widehat{D}_i}\right). \tag{22}$$

The threshold accuracies are then computed as the proportion of pixels satisfying $\delta_i < 1.25^t$ for $t \in \{1, 2, 3\}$:

$$\delta_t = \frac{1}{N}\sum_{i=1}^{N} \mathbb{I}(\delta_i < 1.25^t), \quad t = 1, 2, 3, \tag{23}$$

where N is the number of valid pixels and $\mathbb{I}(\cdot)$ is the indicator function. These metrics quantify inlier rates under increasingly relaxed thresholds and complement error-based measures.

b. Error metrics. To quantify the numerical discrepancy between predicted and GT depths, we report three standard error measures: absolute relative error ϵ_{ARE}, root mean squared error ϵ_{RMSE}, and squared relative error ϵ_{SRE}. For N valid pixels with prediction $\widehat{D}_i$ and GT D_i, these metrics are defined as:

$$\begin{aligned}
\epsilon_{\text{ARE}} &= \frac{1}{N}\sum_{i=1}^{N}\frac{|\widehat{D}_i - D_i|}{D_i}, \\
\epsilon_{\text{RMSE}} &= \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\widehat{D}_i - D_i)^2}, \\
\epsilon_{\text{SRE}} &= \frac{1}{N}\sum_{i=1}^{N}\frac{(\widehat{D}_i - D_i)^2}{D_i}.
\end{aligned} \tag{24}$$

Here, ϵ_{ARE} measures the mean relative deviation, ϵ_{RMSE} penalizes large errors more heavily, and ϵ_{SRE} emphasizes relative inconsistencies, particularly in regions where depth varies locally.

2) Spinal morphology estimation: The SME stage outputs a spine curve map $\hat{M}$ (white curve on a black background). Let M denote the GT map, and let N_s be the number of test samples.

a. Structural Similarity Index Measure (SSIM). We evaluate structural agreement between $\hat{M}$ and M using SSIM:

$$\text{SSIM}(\hat{M}, M) = \frac{(2\mu_{\hat{M}}\mu_M + C_1)(2\sigma_{\hat{M}M} + C_2)}{(\mu_{\hat{M}}^2 + \mu_M^2 + C_1)(\sigma_{\hat{M}}^2 + \sigma_M^2 + C_2)}, \tag{25}$$

and report the dataset mean:

$$\overline{\text{SSIM}} = \frac{1}{N_s}\sum_{n=1}^{N_s}\text{SSIM}(\hat{M}^{(n)}, M^{(n)}), \tag{26}$$

where $\mu_{\hat{M}}$ and μ_M are pixel-wise means, $\sigma_{\hat{M}}^2$ and σ_M^2 are variances, N_s is the number of samples, and $\sigma_{\hat{M}M}$ is the covariance between $\hat{M}$ and M. C_1 and C_2 are small stabilization constants to avoid numerical instability when the denominator approaches zero. All statistics are computed at the pixel level for consistency across samples.

b. Mean length difference $\Delta\bar{\mathcal{L}}$. We extract a centerline from each map and resample it to K ordered points. Let $\Gamma = \{p_i\}_{i=1}^K$ and $\hat{\Gamma} = \{\hat{p}_i\}_{i=1}^K$ denote the GT and predicted centerlines, respectively. The polyline length is defined as:

$$\mathcal{L}(\Gamma) = \sum_{i=1}^{K-1}\|p_{i+1} - p_i\|_2, \quad \mathcal{L}(\hat{\Gamma}) = \sum_{i=1}^{K-1}\|\hat{p}_{i+1} - \hat{p}_i\|_2, \tag{27}$$

and the per-sample length error $\Delta\mathcal{L}^{(n)}$ and the dataset mean $\Delta\bar{\mathcal{L}}$ are:

$$\Delta\mathcal{L}^{(n)} = |\mathcal{L}(\hat{\Gamma}^{(n)}) - \mathcal{L}(\Gamma^{(n)})|, \quad \Delta\bar{\mathcal{L}} = \frac{1}{N_s}\sum_{n=1}^{N_s}\Delta\mathcal{L}^{(n)}. \tag{28}$$

Here, p_i and $\hat{p}_i$ denote the i-th sampling point on the GT and predicted centerlines, respectively. K denotes the number of resampled points, $\mathcal{L}(\cdot)$ denotes the centerline length calculation operator, and N_s denotes the total number of samples in the dataset.

c. Mean curvature difference $\Delta\bar{\kappa}$. To quantify shape consistency, we compute the discrete curvature along the resampled centerline. For interior points $i = 2, \ldots, K-1$, we define the adjacent segment vectors:

$$v_i^- = p_i - p_{i-1}, \quad v_i^+ = p_{i+1} - p_i, \tag{29}$$

and the turning angle θ_i using the cosine similarity:

$$\theta_i = \arccos\left(\frac{\langle v_i^-, v_i^+\rangle}{\|v_i^-\|_2\|v_i^+\|_2}\right). \tag{30}$$

The local arc-length scale is approximated by the average chord length $l_i = \frac{1}{2}(\|v_i^-\|_2 + \|v_i^+\|_2)$, and the discrete curvature is defined as:

$$\kappa_i = \frac{\theta_i}{l_i}. \tag{31}$$

Let $\hat{\kappa}_i$ denote the curvature computed from the predicted centerline $\hat{\Gamma}$. The per-sample curvature error $\Delta\kappa^{(n)}$ and the dataset mean $\Delta\bar{\kappa}$ are:

$$\begin{aligned}
\Delta\kappa^{(n)} &= \frac{1}{K-2}\sum_{i=2}^{K-1}\left|\hat{\kappa}_i^{(n)} - \kappa_i^{(n)}\right|, \\
\Delta\bar{\kappa} &= \frac{1}{N_s}\sum_{n=1}^{N_s}\Delta\kappa^{(n)}.
\end{aligned} \tag{32}$$

Together, $\overline{\text{SSIM}}$, $\Delta\bar{\mathcal{L}}$, and $\Delta\bar{\kappa}$ evaluate agreement in appearance, global extent, and local geometry, respectively.

C. Experimental settings

To improve robustness and increase effective training diversity, we applied data augmentation during model training. Augmentations included random horizontal flipping (probability = 0.5), random vertical flipping (probability = 0.3), and additive Gaussian noise (mean = 0, standard deviation = 0.02). For components that rely on patch partitioning and reassembly (e.g., the BSDE module), geometric transformations were applied in a manner that preserves patch topology and spatial correspondence, so as not to compromise inter-patch continuity and depth consistency. For consistency across experiments,

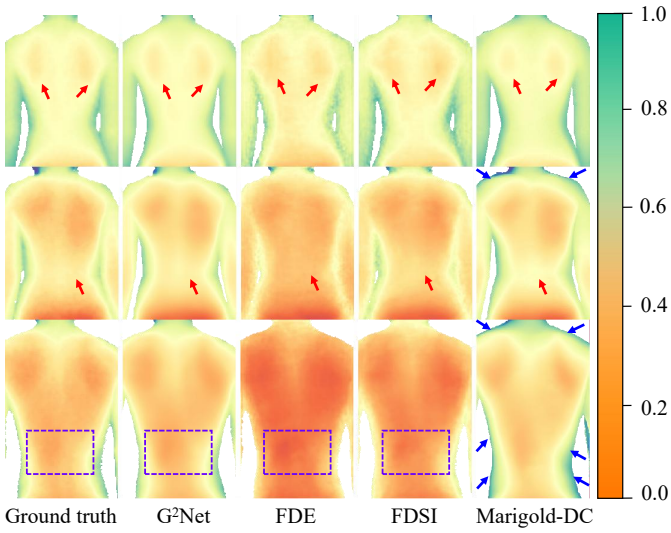


Fig. 7: Visual comparison of back-surface depth estimation results. Columns (from left to right) display the GT, G^{2}Net, FDE [56], FDSI [57], and Marigold-DC [58]. Rows correspond to different cases. Red arrows highlight depth transition quality, blue arrows indicate depth estimation performance near the back boundary, and purple boxes mark regions with pronounced surface undulations.

all input images were resized to 480×240 prior to training and inference. All models were implemented in PyTorch and trained on an NVIDIA RTX 5080. The initial learning rate was set to 1×10^{-4} for BSDE and 1×10^{-5} for SME, with a batch size of 2 for both stages. Optimization was performed using an adaptive moment estimation optimizer (AdapMEM) to minimize the corresponding training objectives.

V. Results

To comprehensively evaluate the proposed G^{2}Net for radiation-free spinal morphology assessment, we report experimental results from its two core stages, namely BSDE and SME. For each stage, both quantitative metrics and qualitative visual comparisons are provided to assess numerical accuracy and structural fidelity, respectively. To ensure a fair and reproducible comparison, all baseline methods were implemented under identical experimental conditions, including the same preprocessing pipeline and unified training settings (e.g., batch size, learning rate, and optimizer), unless otherwise specified.

A. Back-surface depth estimation

We compare our model against three representative monocular depth estimation approaches: FDE [56], FDSI [57], and Marigold-DC [58]. These baselines are all representative algorithms that have been widely verified in the current field of monocular depth estimation and can reflect diverse design philosophies, spanning geometry-inspired modeling, deep learning feature fusion and multimodal depth estimation. Thus, they provide meaningful references for assessing performance on fine-scale back-surface depth.

TABLE I: Quantitative Comparison of Monocular Depth Estimation Models

Model	$\epsilon_{\text{ARE}}\downarrow$	$\epsilon_{\text{RMSE}}\downarrow$	$\epsilon_{\text{SRE}}\downarrow$	$\delta_1\uparrow$	$\delta_2\uparrow$	$\delta_3\uparrow$
FDE [56]	0.204	0.129	0.046	0.658	0.877	0.960
FDSI [57]	0.142	0.091	0.028	0.805	0.944	0.985
Marigold-DC [58]	0.193	0.088	0.084	0.930	0.977	0.981
G^{2}Net	**0.053**	**0.048**	**0.006**	**0.962**	**0.990**	**0.993**

Table I reports depth estimation performance using the metrics defined in IV-B.1 G^{2}Net achieves the best performance across all metrics, yielding the lowest errors ($\epsilon_{\text{ARE}} = 0.053$, $\epsilon_{\text{RMSE}} = 0.048$, $\epsilon_{\text{SRE}} = 0.006$) and the highest threshold accuracies ($\delta_1 = 0.962$, $\delta_2 = 0.990$, $\delta_3 = 0.993$). These results indicate that G^{2}Net provides both accurate absolute depth estimates (low error) and high multiplicative consistency with GT (high δ scores). In comparison, FDE exhibits the highest errors across all metrics and the lowest δ scores, revealing its limited capability in modeling fine-grained surface geometry. FDSI performs moderately better but still substantially behind G^{2}Net across all metrics. Marigold-DC achieves a competitive ϵ_{RMSE} but shows higher relative errors ($\epsilon_{\text{ARE}}/\epsilon_{\text{SRE}}$) and , along with noticeably weaker δ scores, indicating reduced proportional depth consistency. Overall, G^{2}Net establishes itself as the most reliable solution for back-surface depth estimation, providing a strong foundation for subsequent spinal morphology inference.

Fig. 7 illustrates the qualitative comparison of predicted back-surface depth maps across different models. The heatmaps clearly demonstrate that G^{2}Net produces depth predictions that most closely align with the GT, showing smooth gradients, well-preserved spatial structures, and anatomically coherent depth transitions (red arrows in Fig. 7).

In contrast, FDE suffers from depth predictions and distorted structural patterns, with smooth depth transitions. FDSI improves local details relative to FDE, but the predictions remain overly smoothed and still deviate substantially from the GT, particularly in regions with pronounced surface undulations where fine-scale variations are not faithfully recovered (purple boxes in Fig. 7). Marigold-DC generates plausible depth estimates in the central back region, yet performs poorly near the body boundary (blue arrows in Fig. 7), exhibiting noticeable deviations and inconsistent depth distributions. Overall, these visual results corroborate the quantitative findings and demonstrate the superior representational capacity and structural fidelity of G^{2}Net for back-surface depth estimation.

B. Spine morphology estimation

To evaluate the effectiveness of G^{2}Net in spinal morphology estimation, we quantitatively and qualitatively compare our model against several classical generative models: ControlNet [48] and pix2pix [59]. The goal is to assess how well each model captures the anatomical characteristics of the spine from back-surface images.

Table II reports the quantitative results under the three evaluation metrics defined in Sec. IV-B.2. G^{2}Net achieves the best overall performance, with the highest $\overline{\text{SSIM}}$ (0.950 ± 0.013), indicating superior structural consistency with the GT. It also records the smallest mean length difference (0.079)

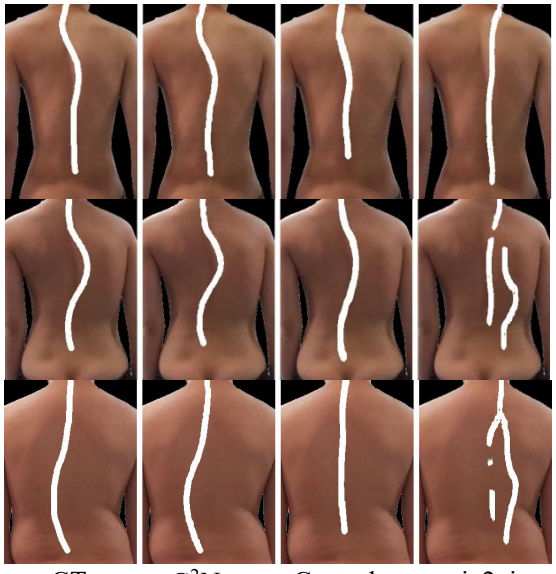


Fig. 8: Visual comparison of spinal morphology estimation using different models. Rows (from left to right) show the GT, the results of G^{2}Net, ControlNet [48], and pix2pix [59], respectively. The morphology of the spine is represented by continuous white curves.

and curvature difference (0.028), reflecting more accurate and stable spine shape reconstructions. ControlNet performs competitively in terms of $\overline{\text{SSIM}}$ (0.949 ± 0.012) but exhibits larger deviations in both length and curvature, suggesting limitations in preserving geometric precision. In contrast, pix2pix shows the weakest performance, with notable degradation across all metrics, highlighting deficiencies in both global morphology understanding and local contour fidelity. These quantitative findings confirm that the proposed G^{2}Net provides more precise and reliable estimates of spinal morphology, especially in preserving fine curvature and spatial structure.

Fig. 8 illustrates the qualitative comparison of spinal contour predictions from different models, overlaid on the original RGB back images. The white curves denote the GT and estimated spinal shapes. G^{2}Net yields the most anatomically coherent spinal curves, closely matching the GT across various spinal conditions.

Notably, in moderate-to-severe deformities and cases with pronounced postural variations, the predicted curves preserve curvature magnitude, endpoint localization, and global continuity, thereby faithfully recovering the spinal morphology and its spatial configuration.

In contrast, ControlNet generally captures the coarse trajectory of the spine but often produces less precise contours with reduced smoothness and weaker adaptation to subtle curvature variations. pix2pix frequently yields fragmented or distorted curves and exhibits noticeable spatial misalignment, failing to reliably capture the underlying spinal structure. Overall, both qualitative and quantitative results consistently demonstrate that G^{2}Net provides more anatomically faithful and geometrically accurate spinal morphology estimation than the compared baselines.

TABLE II: Performance Comparison of Spinal Morphology Estimation Models

Model	SSIM↑		Length diff.↓		Curvature diff.↓	
	mean	SD	mean	SD	mean	SD
ControlNet [48]	0.949	0.012	0.131	0.212	0.051	0.175
pix2pix [59]	0.914	0.015	0.196	0.167	0.030	0.021
G^{2}Net	**0.950**	**0.013**	**0.079**	**0.148**	**0.028**	**0.112**

C. Ablation study

To evaluate the contribution of each core component in G^{2}Net, we define the baseline model as the complete architecture that integrates four key modules. These modules include the ConvNeXt image encoder, the Mamba-based feature integrator, the cross-scale feature fusion module, and the cascade upsampling decoder. This baseline model serves as the performance reference for assessing the effect of removing individual components.

We then conduct a comprehensive ablation study by removing each module individually while keeping all remaining components, training configurations such as data augmentation, batch size, and learning rate, as well as evaluation settings, consistent with the baseline. For each ablated variant, we report quantitative results measured by absolute relative error e_{ARE}, root mean squared error e_{RMSE}, squared relative error e_{SRE}, and threshold accuracies δ_1, δ_2, and δ_3. In addition, qualitative visual comparisons of back-surface depth predictions are provided to verify the role of each module in maintaining high-precision depth estimation performance.

Table III summarizes the quantitative evaluation of each ablation variant using the error and accuracy metrics defined in Section III-A. The full G^{2}Net model achieves the best performance across all metrics, with minimal absolute relative error ($\epsilon_{\text{ARE}} = 0.158$), RMSE ($\epsilon_{\text{RMSE}} = 0.060$), and square relative error ($\epsilon_{\text{SRE}} = 0.011$), along with the highest threshold accuracies ($\delta_1 = 0.821$, $\delta_2 = 0.940$, $\delta_3 = 0.976$). Notably, excluding the ConvNeXt image encoder, Mamba-based feature integrator, or the cross-scale feature fusion module results in substantial increases in all error metrics and a corresponding decline in δ thresholds. Although the model without the cascade decoder achieves lower RMSE and SRE than some other variants, its absolute error remains high and threshold accuracies are significantly reduced—highlighting the decoder's indispensable role in refining final output accuracy. Collectively, these results underscore the synergistic contribution of all components in G^{2}Net. Their integration is vital for robust, high-precision back-surface depth estimation and reliable downstream spinal morphology analysis.

Fig. 9 presents the visual comparison of back-surface depth predictions across different ablated versions of G^{2}Net. The complete G^{2}Net model produces predictions that are most consistent with the GT, capturing fine-grained depth transitions across the trunk region. Removing the image encoder substantially degrades the core depth representation, resulting in less accurate predictions and pronounced deviations near anatomical boundaries. Eliminating the Mamba-based

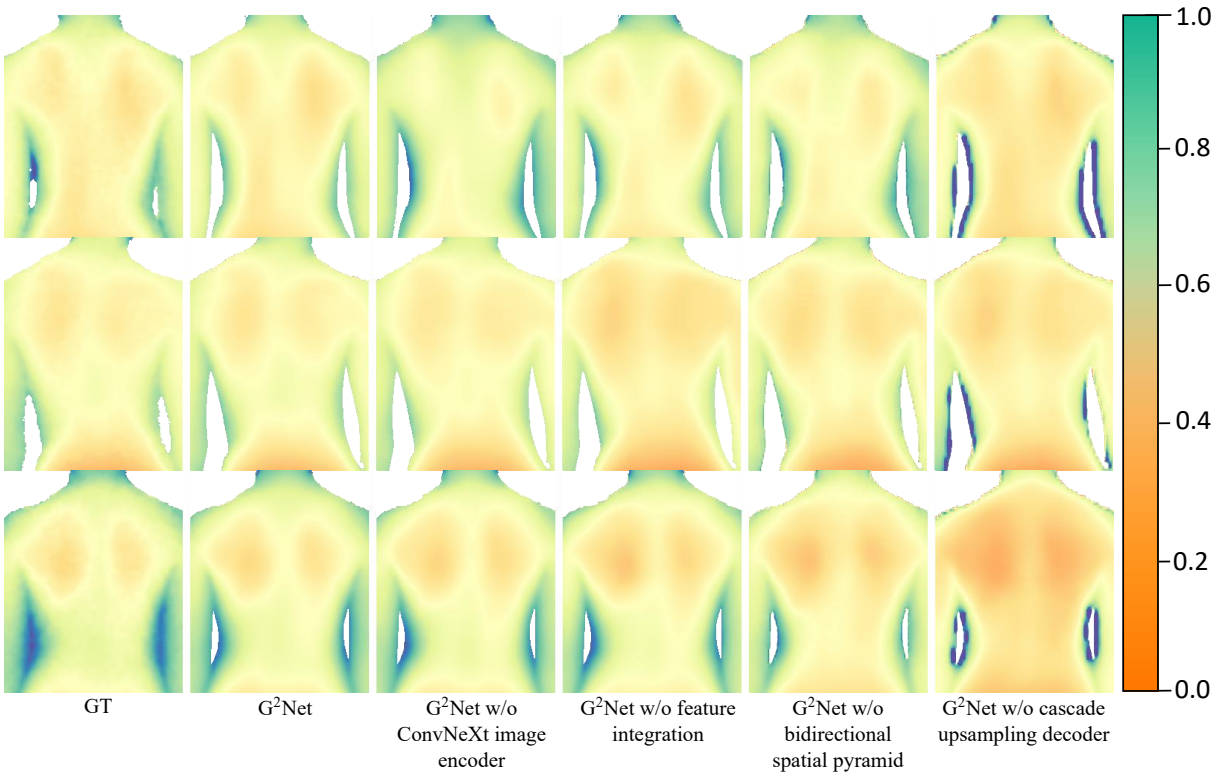


Fig. 9: Visualization comparison of the ablation study for human back-surface depth estimation. Columns (from left to right) show the GT, the full model, the model without ConvNeXt image encoder, without feature integrator, without feature fusion module, and without cascade upsampling decoder. Depth values are encoded by the color intensity of the heatmaps.

TABLE III: Ablation Study on the Components of our model

Model Variant	Components				Metrics					
	ConvNeXt image encoder	Mamba-based feature integrator	cross-scale feature fusion module	cascade upsampling decoder	$\epsilon_{\text{ARE}} \downarrow$	$\epsilon_{\text{RMSE}} \downarrow$	$\epsilon_{\text{SRE}} \downarrow$	$\delta_1 \uparrow$	$\delta_2 \uparrow$	$\delta_3 \uparrow$
✓ denotes enabled; ✗ denotes removed.										
w/o ConvNeXt image encoder	✗	✓	✓	✓	0.690	0.251	0.332	0.538	0.702	0.788
w/o Mamba-based feature integrator	✓	✗	✓	✓	0.714	0.255	0.342	0.531	0.695	0.782
w/o cross-scale feature fusion module	✓	✓	✗	✓	0.698	0.256	0.334	0.524	0.690	0.778
w/o cascade upsampling decoder	✓	✓	✓	✗	0.707	0.182	0.166	0.558	0.730	0.796
G²Net (full model)	✓	✓	✓	✓	**0.158**	**0.060**	**0.011**	**0.821**	**0.940**	**0.976**

feature integrator leads to discontinuities in depth transitions and blurred representations in localized regions, especially in the lumbar area, underscoring the module's role in cross-level feature integration. Omitting the cross-scale feature fusion module compromises depth consistency across scales, demonstrating its critical role in enhancing semantic-geometric synergy. The variant without the cascade upsampling decoder yields the most noticeable artifacts and misaligned depth predictions in both the back and lumbar regions, revealing the decoder's importance in progressive resolution recovery and contour fidelity. These observations confirm that all proposed modules are essential for capturing accurate depth structure and ensuring high-fidelity geometric estimation.

VI. Discussion

This study introduces G²Net, a two-stage, radiation-free framework for spinal morphology assessment from a single back-surface RGB image. Across both stages, the proposed design consistently improved (i) high-precision back-surface depth recovery and (ii) spine curve generation that aligns with X-ray–derived morphology, supporting the premise that explicit geometric priors can substantially strengthen optical AIS assessment under practical imaging variability.

A. Interpretation of the main findings

In the back-surface depth estimation stage, G²Net achieved lower depth errors and higher threshold accuracies than several representative baselines. This performance indicates improved sensitivity to subtle surface undulations that current MDE methods often suppress. By combining global spatial continuity with patch-wise local cues, our design effectively minimizes the trade-off between overregularized smoothness and discontinuous reconstructions. This is further enhanced by our

Mamba-based integration strategy, which efficiently captures long-range dependencies to preserve global consistency at high spatial resolutions [60].

For spinal morphology estimation, conditioning the generative model on both RGB appearance and estimated depth resulted in spine curve maps with higher structural fidelity. This finding aligns with the clinical intuition that surface texture alone is an incomplete proxy for underlying 3D anatomy. Depth information provides essential geometric constraints that regularize the inferred spinal trajectory and reduce geometric discrepancies.

B. Clinical implications

From a translational perspective, the proposed framework offers two key practical advantages. First, inference requires only a single RGB image, without the need for depth input during deployment, thereby simplifying and streamlining image acquisition in clinical settings. Second, the generated spine curve map serves as a compact yet informative intermediate representation that can be readily integrated into downstream clinical workflows. Examples include automated extraction of spinal curvature descriptors, longitudinal monitoring of morphological changes, and triage decision support. In settings where repeated radiographs are undesirable (e.g., AIS follow-up), such a tool could support more frequent monitoring and flag cases needing confirmatory imaging, rather than aiming to replace radiography outright.

C. Limitation and future directions

Current results show strong performance on the collected cohort but lack sufficient generalizability, as robustness to new sites, cameras, lighting, body habitus and pose remains unvalidated through multi-center external testing. Additionally, the real spine curve map derived from X-ray landmark annotations and subsequent interpolation or smoothing may introduce bias toward specific curve characteristics.

Future work will focus on three key aspects. It will prioritize broader clinical validation and calibration across different devices and sites. It will also strive to directly estimate clinically actionable metrics such as Cobb-related measurements from the generated curve, along with uncertainty quantification. Moreover, it will enhance robustness under unconstrained capture scenarios while preserving the fine-scale geometry that is essential for early-stage AIS assessment.

VII. Conclusion

This study presents G^{2}Net, a novel two-stage, radiation-free framework for spinal morphology assessment using a single RGB image of the back surface. Addressing the clinical need for non-invasive scoliosis evaluation, G^{2}Net integrates a geometry-aware depth estimation module and a conditional generative model to reconstruct anatomically faithful spinal curves. In the first stage, it achieves superior performance across both quantitative metrics and visual consistency, effectively capturing fine-scale back surface undulations critical for downstream analysis. In the second stage, G^{2}Net leverages the estimated depth cues to generate structurally accurate spine curve maps, significantly outperforming existing image-to-image generation models in both fidelity and clinical alignment. The experimental results confirm that explicitly incorporating geometric priors, even under realistic imaging variations, substantially enhances model robustness and anatomical consistency. The proposed approach requires only a single RGB image at inference, facilitating broader clinical adoption in scenarios where repeated radiographic exposure is undesirable.

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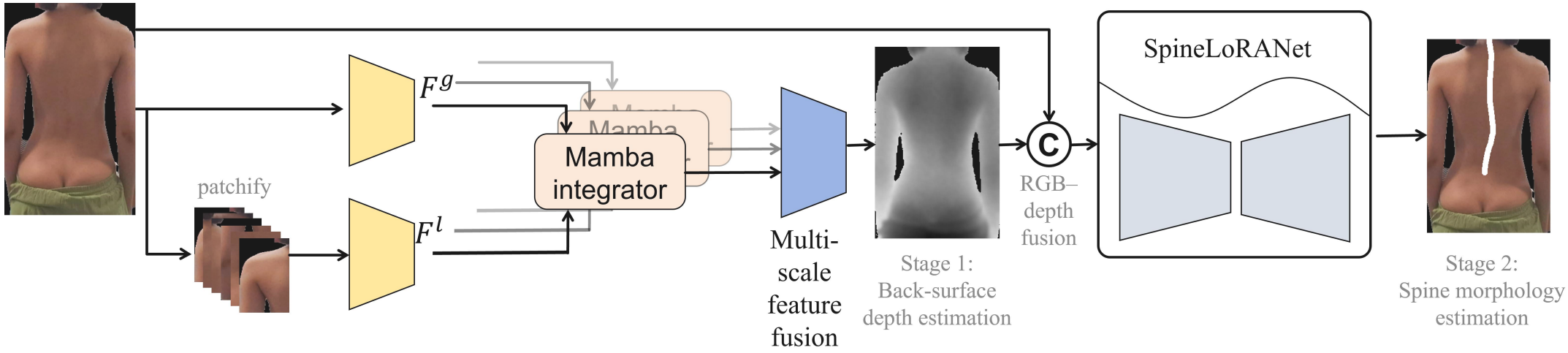
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Multi-scale feature fusion
Stage 1: Back-surface depth estimation
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Stage 2: Spine morphology estimation

G²Net is a geometry-guided, radiation-free framework for AIS assessment from one back RGB image. It predicts back-surface depth and uses RGB–depth conditioning to generate anatomically consistent spine-curve maps for spine morphology estimation.